

Distribution Calculators Guide

How to Use MDragon Distribution Calculators

Normal Distribution Calculator

Inputs Required

- μ (mean): Population/sample mean
- σ (std dev): Standard deviation
- x : Value of interest

What It Calculates

1. **Z-score**: How many SDs from mean
2. **$P(X < x)$** : Probability below x
3. **$P(X > x)$** : Probability above x
4. **Percentile**: What % falls below x

Example

 **Problem:** Heights: $\mu=70"$, $\sigma=3"$. Find $P(X < 73)$.

Steps:

1. Enter $\mu = 70$
2. Enter $\sigma = 3$
3. Enter $x = 73$
4. Read result: $P(X < 73) = 0.8413$ or 84.13%

Binomial Distribution Calculator

Inputs Required

- **n:** Number of trials
- **p:** Probability of success (0 to 1)
- **x:** Number of successes
- **Type:** =, ≤, ≥, <, >

Example

 **Problem:** 10 coin flips, P(exactly 6 heads)?

Steps:

1. $n = 10$
2. $p = 0.5$
3. $x = 6$
4. Type = "Exactly (=)"
5. Result: 0.2051 or 20.51%

Tips

- For "between" use: $P(a \leq X \leq b) = P(X \leq b) - P(X \leq a-1)$
- Check cumulative option for ranges

Poisson Distribution Calculator

Inputs Required

- λ (lambda): Average rate
- **x:** Number of occurrences
- **Type:** =, ≤, ≥, <, >

Example

 **Problem:** 4 emails/hour average. P(exactly 6 in next hour)?

Steps:

1. $\lambda = 4$
2. $x = 6$
3. Type = "Exactly (=)"
4. Result: 0.1042 or 10.42%

Important

- Adjust λ for time interval!
- If given "per hour" but want "per 30 min", divide λ by 2

Hypothesis Test Calculator

Inputs Required

- **Test type:** One-sample, two-sample, paired
- **α :** Significance level (usually 0.05)
- **Tail:** Two-tailed, right, or left
- **Sample statistics:** $\bar{x}$, s , n

Example

 **Problem:** Test if $\mu = 100$ when $\bar{x}=105$, $s=15$, $n=25$, $\alpha=0.05$

Steps:

1. Select "One-sample t-test"
2. Enter $\mu_0 = 100$
3. Enter $\bar{x} = 105$, $s = 15$, $n = 25$
4. Select "Two-tailed"
5. $\alpha = 0.05$

6. Result shows: t-statistic, p-value, decision

Practice with different values to build intuition!

MDragon Data Tools | Statistics Education Platform

mondragon-developer.github.io/statools